



सत्यमेव जयते

NITI Aayog

DRAFT

SECTOR AT A GLANCE

Food Crops

Agriculture and Allied Sectors in India

2023-Agriculture-SAG-03

Sector at a Glance

Food Crops

Agriculture and Allied Sectors in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Food Crops. NITI Aayog Sector-at-a-Glance Series – Agriculture and Allied Sectors in India, 2023-Agriculture-SAG-03, <link>

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Since the advent of the Green Revolution, enhancing the cultivation and yield of food crops has been an integral part of India's agricultural strategy to achieve nationwide food security. Between 1950 and 2022,¹ the production of food crops has increased six-fold with the government focusing heavily on infrastructure investments, irrigation and targeted input-based support for fertilisers and pesticides, particularly for rice and wheat. Concerted policy efforts through the years have resulted in surplus production of some of these dominant crops.

Due to the changing nature of consumer demand and government priorities, there has been a growing shift from food crops to commercial crops and horticultural produce. While cereals constituted a 37.1% share in the value of total agriculture Gross Value Added (GVA) during the 1970s, their contribution declined to 18.5% by 2021. A similar decline was seen in the GVA of pulses although the domestic demand for the crop has been on the rise since 2019-20.

Crops solely cultivated for feeding purposes are referred to as food crops, and primarily consist of two groups: cereals and pulses. India's unique geography and tropical climate support a diverse range of food crops during the seasons of Rabi and Kharif. The analysis and insights in this document will largely focus on these groups as they collectively constitute 99.6% of the GVA contribution of food crops.² The aspects covered involve overarching trends in production, and the area, yield and trade at the aggregate- and crop- specific levels. The document also delves into major challenges, opportunities and developments crucial to determining the way forward for these crops.

Figure 1: Major Food Crops of India

		 Cereals Wheat, rice, maize, finger millet, jowar, bajra	 Pulses Tur, urad, moong, peas, gram, lentils
 Overall Value Contribution (2021-22)	Agricultural GVA (%)	18.5	3.9
	Total GVA (%)	2.7	0.6
 Trade (2021-22)	Exports as share of total production (%)	11.2	1.4
	Imports as share of total domestic consumption (%)	0.5	10
 Production (2017-18 to 2021-22)	Total production in MT ² (2021-22)	288	27.7
	Compound annual growth rate (CAGR) (%) 2017-18 to 2021-22	2	1.7
 Yield (Kg/ha)	2017-18	2,657	853
	2021-22	2,895	892

Source: Estimates based on data from National Accounts Statistics, the Ministry of Statistics and Programme Implementation (MoSPI), and NITI Aayog's Demand and Supply Projections towards 2033 of Crops, Livestock, Fisheries and Agricultural Inputs Working Group Report

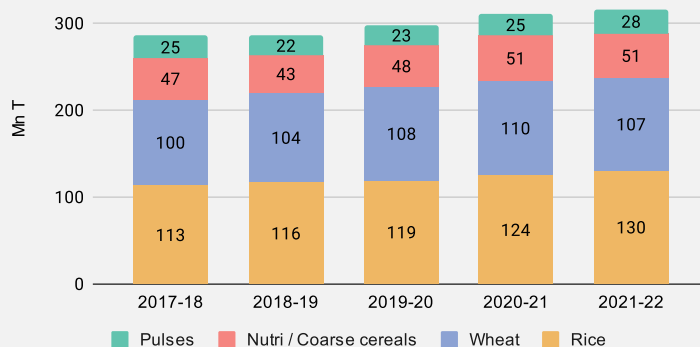
- 1 Ministry of Agriculture & Farmers Welfare. (2022). *Third Advance Estimates*. The Ministry released third advance estimates for the 2022-23 data on production in May 2023. According to those estimates, total production of food crops in 2022-23 was 330.5 MT at a CAGR of 0.92%
- 2 Ministry of Agriculture & Farmers Welfare. (2022). *Agricultural Statistics at a Glance*- This document has been referred to as the primary source for all the crop-specific data on production, area, yield and trade. The 2021-22 figures mentioned throughout the document are based on fourth advance estimates declared by the Ministry of Agriculture on 11.08.2021. In addition, the 2021-22 figures for both exports and imports are projected values.

Production

India is the second-largest producer of food crops, globally. Food crop production has demonstrated an average growth rate of 2.6% in recent years (2017-22) with the total production in 2021-22 amounting to an estimated 315 million tonnes.² Compared to countries like the United States of America, the European Union and China, India still lags behind in terms of food crop yield.

Cereals have been the dominant of the two groups with wheat and rice collectively making up 75% of the total food crops produced in 2021-22. Average growth rates in production of major food crops have been in the range of 1.7% (wheat) and 3.6% (rice). (Figure 2).

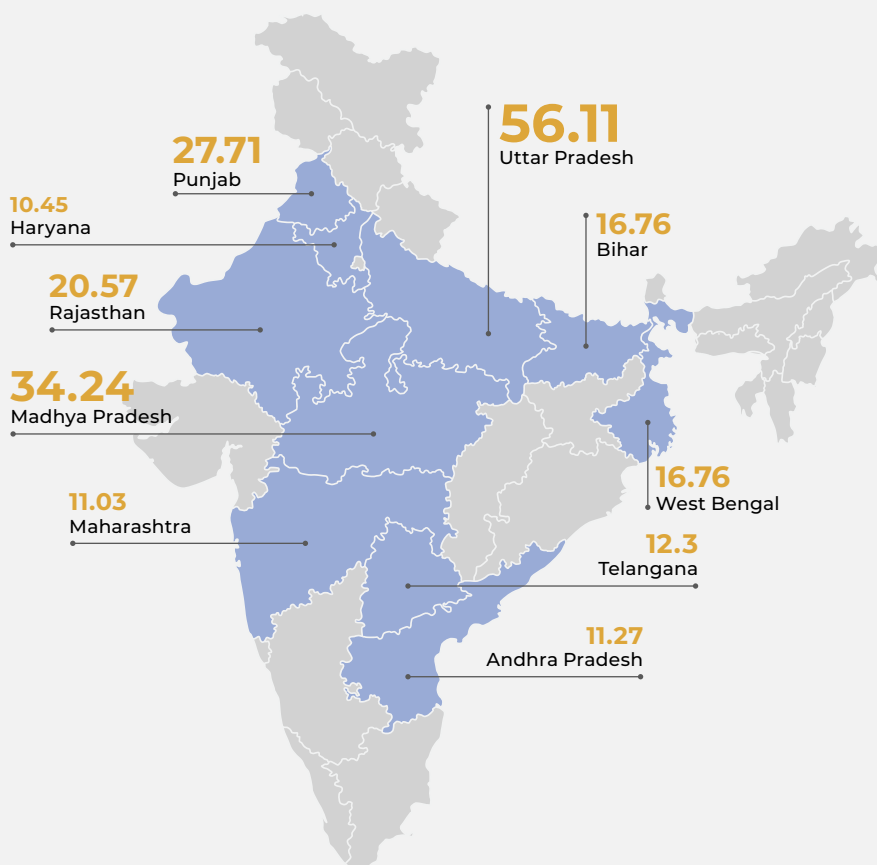
Figure 2: Production Share of Food Crops (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

In 2021-22, cereals accounted for 76.2% (99.5 million ha) of the area under food crop cultivation and pulses covered 23.7% (31 million ha).² Rice and wheat constitute the largest share of food crop production in terms of both aggregate output (75%) and area coverage (58%). Given the dominance of the wheat-rice system in several states and in the Indo-Gangetic plain, various government initiatives in recent years have focused on improving prospects of cultivating pulses and nutri-cereals like millets.

Figure 3: Top 10 Producers of Food Crops in India (Mn T)



Note: The illustrative map only shows the top-producing states as per the cumulative output of rice, wheat, pulses, and nutri-cereals.

Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Domestic and International Market Trends

Demand and consumption patterns are shifting from food crops to commercial crops and horticulture produce. In the food crop sub-sector, the government has been laying more emphasis on improving financial viability and consumption of nutri-cereals and pulses.³



Supply vs Demand Scenario:

Per Capita Production Exceeds Demand Except for Pulses and Millets

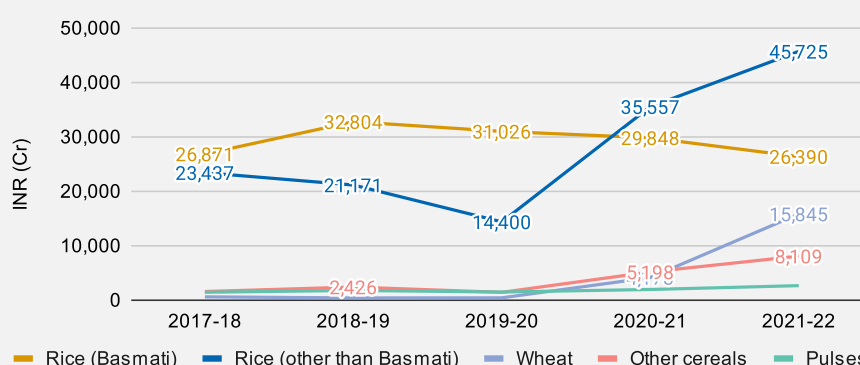
In the last six decades, India has recorded an increase in the per capita growth in the production of all crops except pulses. The per capita production of cereals in the past five decades increased by **80%**. The per capita net availability of cereals in India has been stable at around 168 kilograms for several years now. This is largely in response to shifting market preferences.

The growth in the demand for cereals and pulses from 2020 to 2030 is projected at 21% and 35% respectively, whereas the growth for vegetables and fruits is higher at 68% and 67% respectively.⁴ Although the per capita millet production has declined in recent years, projections suggest that both production and consumption of this crop will increase in the future. The growing prominence of such nutri-cereals is due to the combined effect of a strong thrust in policy and the diversification of consumer diets.

Food crops have experienced a positive CAGR of **12.8%** in trade value from 2017 to 2022, with wheat and nutri-cereals being the leading export commodities.

Food crops constituted **42.6%** of the total agricultural exports in 2021-22 with rice (both Basmati and non-Basmati varieties) making up 73% of the exported food grains. In fact, India's non-Basmati exports have grown by **109%** in the past decade with a steep increase since 2019. Meanwhile, wheat exports have achieved an unprecedented growth of more than **279%** in 2021-22.

Figure 4: Export of Food Crops (2017-18 to 2021-22)



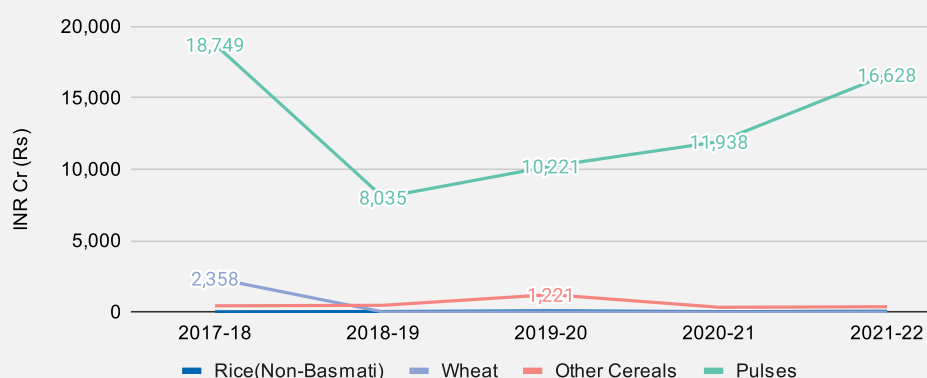
Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Between 2019 and 2022, imports of pulses grew by **27.5%**. As shown in Figure 4, food crops comprised approximately **7.4%** of total agricultural imports with pulses accounting for a substantial **97.5%** of the overall food crop imports. Despite focused efforts, pulse production has stagnated and India remains dependent on imports to meet the rising demand for pulses.

³ Cabinet Committee on Economic Affairs. (2023). *MSP For Rabi Crops*, Retrieved on July 2023

⁴ NITI Aayog. (2018). *Demand & Supply Projections towards 2033*.

Figure 5: Import of Food Crops (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Cereals

Production

Cereal production has shown a slow but consistent increase over the years largely owing to continuous policy benefits and access to high-yielding varieties. Price stability and public procurement have historically favoured cereals over other categories of crops. However, the average growth in cereal production has remained sluggish at 2.6% over the period 2017-18 to 2021-22. The slower growth in cereals has coincided with growing market demand for other commodities such as commercial crops and horticulture produce.⁵ Furthermore, nutri-cereals such as millets have been receiving a conscious policy thrust to improve sustainability and the nutritional security of the nation.

303 MT⁶

Total cereal production (2022-23)

2.6%²

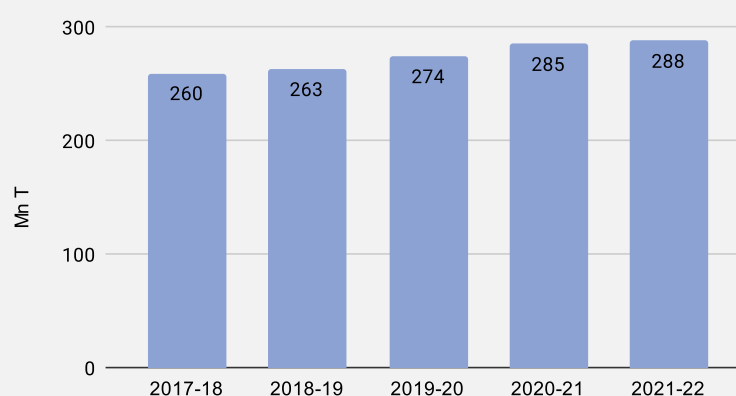
(2017-18 to 2021-22)

90%²

of total food crop production (2021-22)

In terms of compounded growth in production, rice leads at **3.6%** followed by nutri-cereals and wheat at **2.1%** and **1.7%** respectively. In comparison, horticulture crops have achieved a growth rate of 3.5% in the same period in response to consumer preferences.

Figure 6: Production of Cereals (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

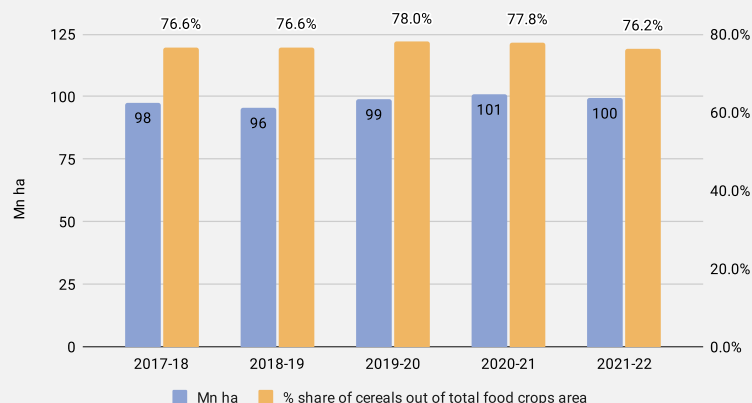
5 R.Chand et al., (2023). From Green Revolution to Amrit Kaal: Lessons and Way Forward for Indian Agriculture.

6 As per third advance estimates, the total production of cereals was 303 MT in 2022-23 at a CAGR of 1.02%.

Area

The absolute area as well as the percentage area under cereal cultivation have been more or less stagnant in recent years. Changing cropping patterns have led to a portion of the cultivation area for cereals being diverted to non-cereal crops.⁵

Figure 7: Area under Cultivation (2017-18 to 2021-22)



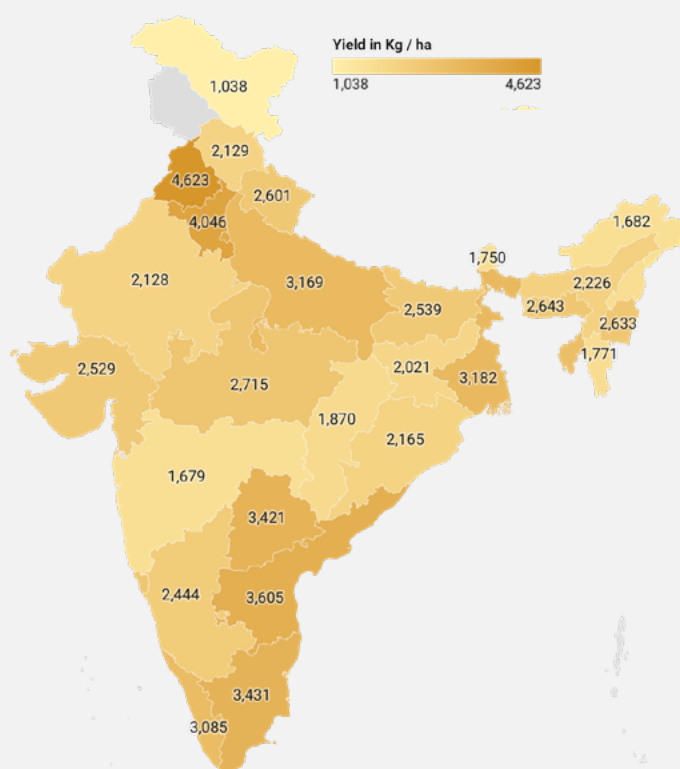
Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Despite a relatively lower growth rate compared to other sub-sectors such as horticulture and dairy, the domestic production of cereals is still growing at a faster rate than domestic demand.⁷

Yield

Cereal yields amounted to 2,895 kilograms per hectare in 2021-22, growing at a CAGR of 2.2% since 2017-18. Among all cereals, wheat offers the highest yield at 3,507 kilograms per hectare while nutri-cereals remain the lowest at 2,247 kilograms per hectare. Maize, one of the major nutri-cereals in India, ranks third after wheat and rice at 2,563 kilograms per hectare. Historically, the yield of cereals has shown a steady improvement due to consistent policy focus on high-quality seeds, irrigation, and improved access to inputs. Punjab and Chandigarh together had the highest cereal yields at 4,600 kilograms per hectare in 2020-21, followed by Haryana and Delhi.

Figure 8: Cereal Yields across States (2020-21)



Source: Ministry of Agriculture and Farmers Welfare

Export

India's cereal exports amounted to INR 96,011 crores in 2021-22,⁸ with rice (including Basmati and non-Basmati varieties) occupying the major share (75% in value terms). India already exports more than 12% of its domestically-produced rice.⁸ The primary export destinations in 2021-22 included Bangladesh, Iran, Saudi Arabia, Nepal, Vietnam, and Benin.⁸

⁷ The expenditure elasticity for cereals in India (a measure of the responsiveness of demand to changes in expenditure on a bundle of similar goods) has been estimated to be negative (-0.10)⁵. The estimate indicates that as Indians raise their expenditure on food in general, their demand for cereals has been on the decline.

⁸ Ministry of Commerce and Industry, Agricultural and Processed Food Exports Development Authority (APEDA). *APEDA Products*, Retrieved on July 2023

Production

India is the second largest producer of rice globally (27% of the total global share) with a consistently high growth in productivity due to government interventions.

135.5 MT⁹

Total rice production (2022-23)

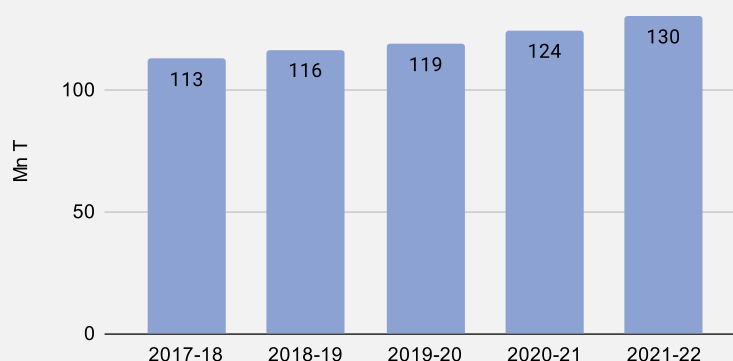
3.6%²

CAGR (2017-18 to 2021-22)

17%²

of total food crop production (2022-23)

Figure 9: Production of Rice (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Major Producers:

West Bengal, Uttar Pradesh, Punjab, Telangana and Odisha comprising **50%** of the total rice production in the country in 2021-22.

Area

The land area under rice cultivation was 46.4 million hectares in 2021-22², growing at a CAGR of 1.5% over the five years since 2017-18. The cultivable area has increased the most for rice compared to all other food crops. However, this area is being diverted to other less water-consuming crops like oilseeds, pulses and nutri-cereal crops. The states using the most area for rice cultivation in 2021-22 included Uttar Pradesh (12.3%), West Bengal (12%), Odisha (8.5%), and Chhattisgarh (8%).

Yield

Rice yields increased to 2,809 kilograms per hectare in 2021-22, growing by 9% since 2017. The introduction of high-yielding varieties, assured irrigation support, fertilisers and pesticides improved rice yields significantly during the past two decades; however, there exists further scope for improvement, such as by incorporating new technologies and adopting hybrid varieties. The states with the highest rice yields in 2021-22 were Punjab, Tamil Nadu, Andhra Pradesh and Telangana.

Cost of Production (A2+FL*)	The national cost of rice production in 2022-23 was INR 1,360 per quintal which was an increase of 16.6% since 2018-19. ²
Demand Supply Estimate	Provided household tastes and preferences remain stable, it is estimated that the demand for rice will increase to 95-105 million tonnes by 2033. ⁴ With an increase in both income and population, it is expected that people will diversify their consumption basket towards fruits, vegetables and high-value crops. Production projections for rice are estimated at 145-151 million tonnes by 2030. ⁴

⁹ As per third advance estimates the total production of rice was 135.5 MT in 2022-23 at a CAGR of 0.8%.

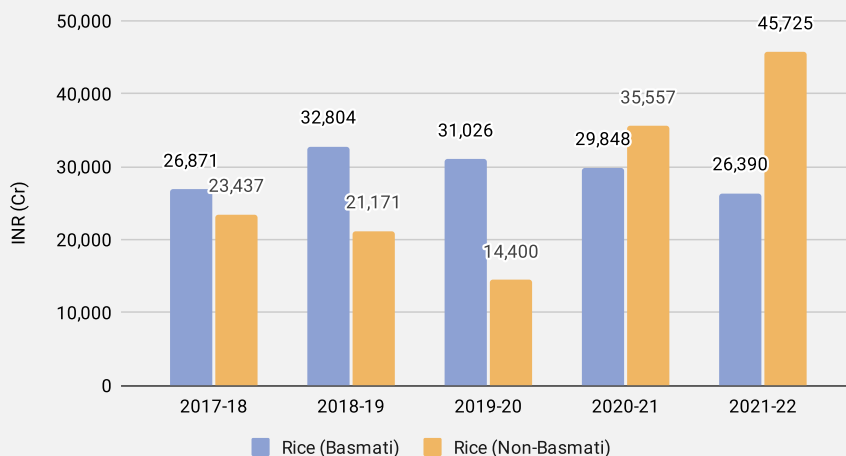
* A2 covers all paid-out costs directly incurred by the farmer in cash and kind on seeds, fertilisers, pesticides, hired labour, leased-in land, fuel, irrigation, etc.. A2+FL includes an imputed value of unpaid family labour.

Export

India accounts for the largest share of the global rice trade, witnessing an upward growth in rice exports of 109% from 2013-14 to 2021-22.⁸

Rice (both Basmati and non-Basmati varieties) occupy the primary share of India's total cereal exports (80% in value terms) during 2022-23. India contributes to nearly **50%** of the world's rice market.⁸

Figure 10: Export of Rice (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Basmati Rice:

India is the leading global exporter of Basmati rice, exporting 3,948,161 metric tonnes of Basmati rice valued at INR 26,416 crores in 2021-22⁸ to its major export destinations: Iran, Saudi Arabia, Iraq, the United Arab Emirates, the United States of America and the Republic of Yemen.⁸

Non-Basmati rice:

The country exported 17,262,235 metric tonnes of non-Basmati rice valued at INR 45,652 crores during 2021-22, where major export destinations included Bangladesh, Benin, China, Nepal, Côte d'Ivoire, Senegal and Togo.⁸

Wheat

Production

India is the second largest wheat producer in the world (14%) and ranks third in yield (3,150 kilograms per hectare), after the United Kingdom and Italy. Since the Green Revolution, the wheat economy has received consistent governmental support with improvements in technology and procurement.

112.7 MT¹⁰

Total wheat production (2022-23)²

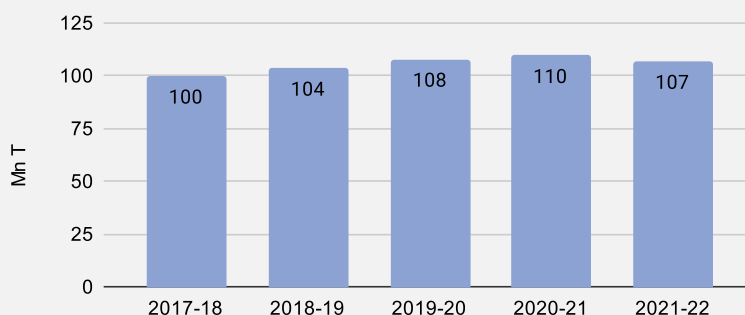
1.7%²

CAGR (2017-18 to 2021-22)

34%²

of total food crop production (2021-22)

Figure 11: Production of Wheat (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

¹⁰ As per third advance estimates released by the Ministry in May 2023, the total wheat production was 112.7 MT in 2022-23 at a CAGR of 1.1%.

Major Producers:

Uttar Pradesh (31.8%), Madhya Pradesh (21%), Punjab (13.9%), Haryana (9.8%) and Rajasthan (8.87%) contributed to **85%** of India's total wheat production in 2021-22.²

Area

The area under wheat cultivation was estimated to be 30.5 million hectares in 2021-22,² growing at a CAGR of 0.68% over five years from 2017-18 to 2021-22. The states with the maximum area under wheat production in 2021-22 include Uttar Pradesh (30.9%), Madhya Pradesh (21.3%), Punjab (14.8%) and Rajasthan (8.5%).

Yield

Wheat yields amounted to 3,507 kilograms per hectare in 2021-22, growing by 4% since 2017-18. The states with the highest yields in 2021-22 included Punjab, Tamil Nadu, Andhra Pradesh and Telangana.

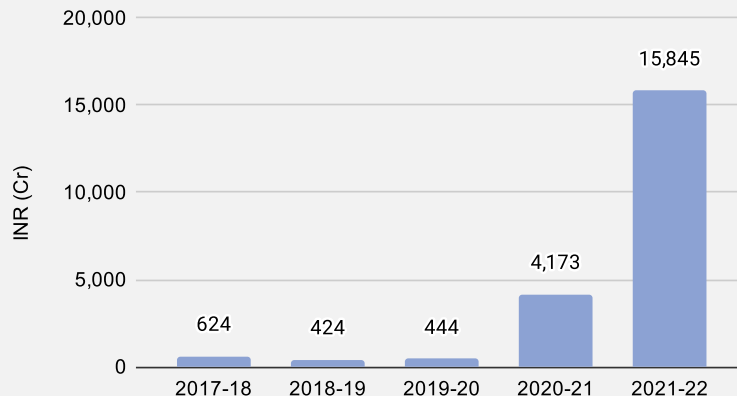
Cost of Production (A2+FL)	The national cost of wheat production in 2022-23 was INR 1,065 per quintal, increasing by 23% over the past five years. ²
Demand Supply Estimate	The export demand for Indian wheat shows an upward trend, projected to reach 113-139 metric tonnes in 2030, while the projected supply is 112-128 metric tonnes for the same year. This burgeoning demand in the international markets can be met by further increasing wheat production. ⁴

Export

India's wheat exports have undergone an immense growth, increasing at a CAGR of 124% from 2017-18 to 2021-22.²

The global wheat trade is larger than that of all other crops combined. India exported 7.2 million metric tonnes of wheat valued at INR 15,840 crores during 2021-22,⁸ to major export destinations such as Bangladesh, Sri Lanka, the United Arab Emirates, the Republic of Yemen, Philippines, and Indonesia.⁸

Figure 12: Export of Wheat (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Nutri Cereals

Production

Nutri or coarse cereals are small-seeded crops which are broadly divided into three categories: major cereals (e.g. Sorghum, finger millet), minor cereals (e.g. Foxtail) and pseudo cereals (e.g. Buckwheat). Millets have low water requirements and offer high nutritional value. In 2020, India produced an estimated 28 metric tonnes of millets.⁸ Nutri or coarse cereals also include maize, which has the highest genetic yield potential of all cereals. As per the Food and Agriculture Organization, India was the fifth largest global producer of maize in 2020 and produced nearly **35.9 metric tonnes** in 2022-23.⁸

India is one of the largest millet producers globally and has taken efforts to make this cereal group an integral part of food security policies. Even though the millet production is increasing, it remains considerably low compared to dominant cereals like wheat and rice.

54.8 MT¹¹

Total coarse cereal production
(2022-23)⁵

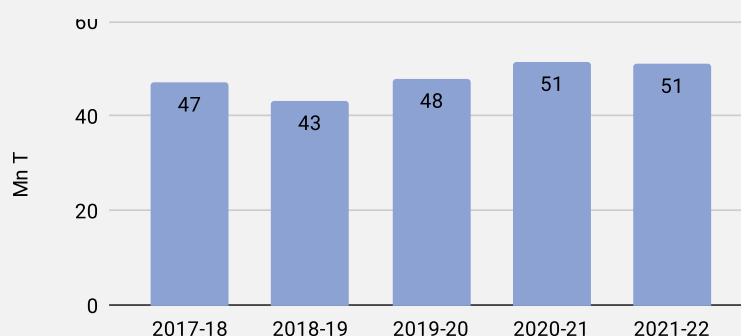
2.1%²

CAGR (2017-18 to 2021-22)

17%²

of total food crop production (2022-23)

Figure 13: Production of Nutri /Coarse Cereals
(2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Major Producers:

Karnataka (14.3%), Rajasthan (13.9%), Maharashtra (11.5%), Madhya Pradesh (11.4%), and Uttar Pradesh (8.5%) contributed to **59.6%** of India's coarse cereal production in 2021-22.

Area

The total area used for cultivating coarse cereals was estimated to be **22.7** million hectares in 2021-22,² decreasing by **7.24%** since 2017. The area used for nutri-cereal cultivation has witnessed a steady decline since the onset of the Green Revolution, with that used for millets experiencing a fall from 45.95 million hectares to 23.83 million hectares between 1970 and 2020.

States with the largest area under coarse cereals in 2021-22:

Rajasthan (24.4%), Maharashtra (16%), Karnataka (14.3%) and Uttar Pradesh (8.8%).

Yield

Coarse cereal yields have increased by **16.2%** since 2017 and were estimated to amount to 2,247 kilograms per hectare in 2021-22.² The states with the highest coarse cereal yields in 2021-22 were West Bengal, Andhra Pradesh, Telangana, and Bihar.

Cost of Production (A2+FL)	The national production cost of certain nutri-cereals like Jowar, Bajra, maize, Ragi and barley ranged between INR 1,000-2,400 per quintal in 2022-23. ² The production cost of these nutri-cereals increased by 22.8% from 2017 to 2022.
Demand Supply Estimate	The national demand for coarse cereals is projected to reach 47.2 million tonnes in 2030 while the projected supply is 64.2 million tonnes for the same year, thereby creating a surplus of 17 million tonnes. This amount is more than double the surplus of coarse cereals in 2020 (7.9 MT). ⁴

¹¹ As per third advance estimates released by the Ministry in May 2023, the total nutri-cereals production was 54.8 MT in 2022-23 at a CAGR of 1.5%.

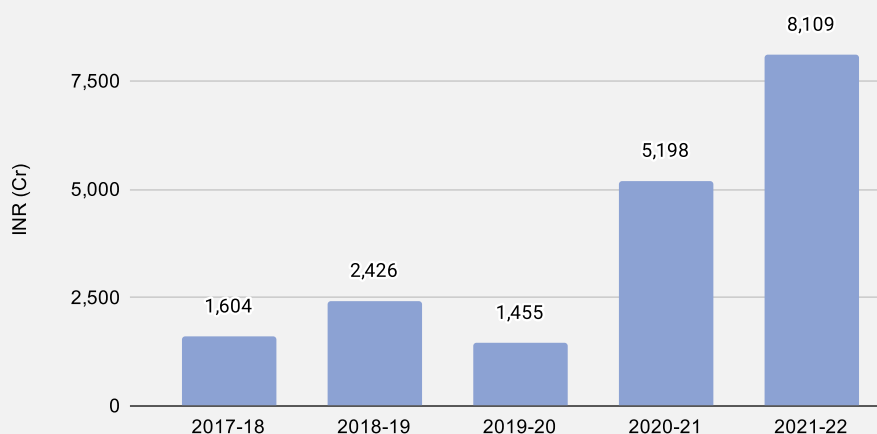
Export

The export footprint of coarse cereals has shown an increase of ~400% between 2017-18 and 2021-22.²

The country exported coarse cereals worth INR 8,109 crores in 2021-22, contributing to 2.2% of its total agricultural exports.⁸

The major export destinations in 2021-22 included the United Arab Emirates, Saudi Arabia, Nepal, the United States of America and Kenya.⁸

Figure 14: Export of Coarse Cereals (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Pulses

Production

Some of the major pulses produced in India include grams, Tur, and lentils. In contrast to cereals, pulse production in India has been stagnant over the years and concentrated only in a few states in the Indo-Gangetic Plain, the coastal south, and the West Bengal delta region. The country depends on imports to meet 10% of its pulse consumption demand. In India, pulses are primarily grown with limited irrigation and quality seeds, which act as a major constraint for farmers. In addition, the relative profitability associated with pulses is less compared to other more remunerative food crops. Therefore, domestic pulse production requires additional long-term policy support to reduce import dependencies.¹²

27.5 MT¹³

Total production in (2022-23)

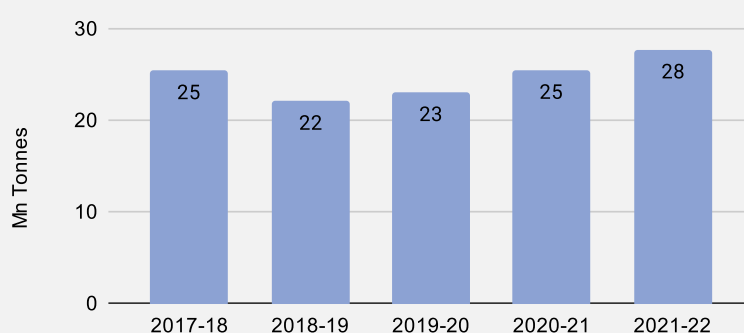
2.9%²

CAGR (FY'18-FY'23)

8%²

of total food crop production (2021-22)

Figure 15: Production of Pulses (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance, 2022 (Ministry of Agriculture and Farmer Welfare)

Major Producers:

Madhya Pradesh, Maharashtra, Rajasthan and Gujarat, comprising **64.8%** of the pulses production in India in 2021-22²

¹² R Kumar. (2018). *Leveraging Policies for Self-Sufficiency of Pulses in India*.

¹³ As per third advance estimates, the total production of pulses was 27.5 in 2022-23 at a CAGR of -0.13%.

India is the largest global producer of pulses followed by Canada and Myanmar. Madhya Pradesh is the top producer of grams and lentils while Maharashtra is the leading producer of Tur.

Area

The total area under cultivation of pulses was estimated to be 31 million hectares in 2021-22,² The area has grown at a CAGR of 1.0% over five years from 2017-18 to 2021-22. Pulses are grown in both Rabi and Kharif seasons and account for 20% of the total area used for food crop cultivation. The states with the largest area for pulse cultivation in 2021-22 were Madhya Pradesh (21.8%), Maharashtra (18.7%), Rajasthan (14.5%), Gujarat (9.71%), and Uttar Pradesh (9.25%).²

Yield

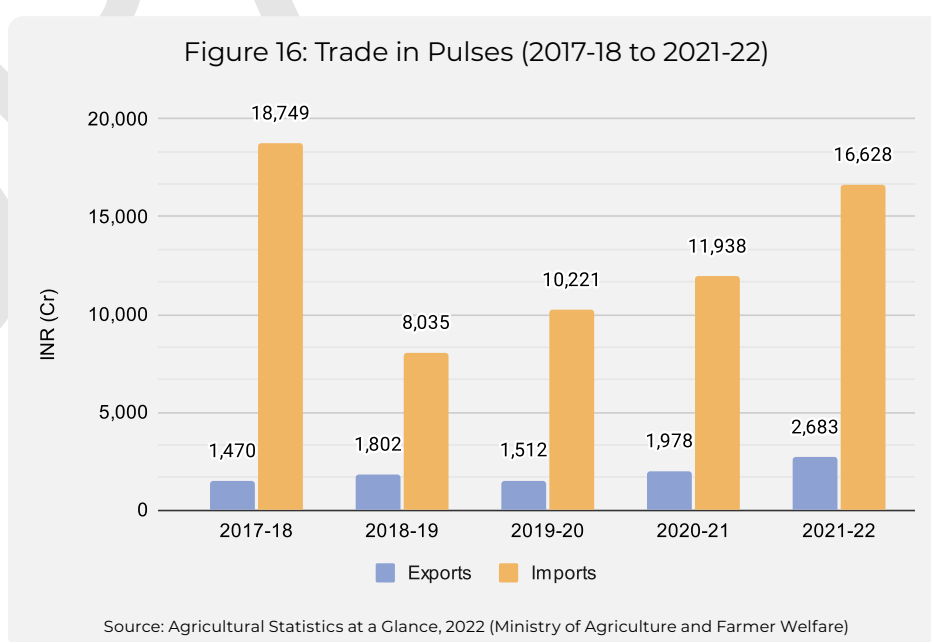
Pulse yields are estimated to be 892 kilograms per hectare in 2021-22, having increased by 4.6% since 2017.* The states with the highest pulse yields in 2021-22 included West Bengal, Andhra Pradesh, Telangana, and Bihar.

Cost of Production (A2+FL)	The national cost of production of pulses like Arhar, Moong, Urad, grams and Masoor in 2022-23 ranges between INR 3,200-5,170 per quintal. This cost has risen by 19.23% since 2017. ²
Demand Supply Estimate	The national demand for pulses is projected to reach 26.6 million tonnes in 2030 while the supply projections for the same year are 26.4 million tonnes; this would create a shortage of 0.2 million tonnes. ⁴ Considering the rising demand for pulses, the government has stated that the domestic pulse economy needs to be strengthened to reduce import dependency to 3% by 2030.

Exports vs Imports: A Comparative Analysis

The growing domestic and global demand for pulses is amplifying the national-level policy focus on reducing import dependency and boosting local production.

While India's pulse exports amounted to 0.71% of its total agricultural export, India's pulse imports made up 7.71% (worth INR 16,627 crores) of the total agricultural imports in 2021-22. However, the export potential for pulses is growing due to the demand for chickpeas and lentils in neighbouring countries such as China and Bangladesh.⁸



India's major export destinations for pulses include Bangladesh, China, the United Arab Emirates, the United States of America and Nepal.⁸



Growing industrial demand for millet and millet products

- India contributes to nearly 19% of the global millet production and estimates suggest that diverting 0.56 million metric tonnes of millets into the export of grain- and millet-based value-added products would yield an export opportunity worth INR 16,000 crores by 2030.¹⁴
- With government initiatives like the International Year of Millets, there is a growing willingness among private-sector brands to expand their product baskets, promote healthier alternatives and leverage policy support for millet cultivation. Established fast-moving consumer goods (FMCG) brands and agribusinesses are including millets like Sorghum and Bajra innovatively in product recipes and food items like cookies and flour.¹⁵



Reducing import dependency for pulses through research and innovation

- Pulse consumption is likely to increase given the rise in population and a growing awareness of the crop's nutritional benefits. Estimates suggest pulse production must grow by at least 2% to meet projected demands.¹⁶ In order to meet the growing domestic and international demand, India can leverage its access to advanced genomic technologies to foster the development of region-specific hybrids that can help bridge existing yield gaps across different geographical regions.¹⁷
- The Indian Institute of Pulses Research (IIPR), in collaboration with the Indian Council of Agricultural Research (ICAR), has made significant achievements in developing high-yielding cultivars like chickpeas and lentils. These innovations need to be reinforced with research into cost-effective seed storage methods to minimise post-harvest loss and inclusive value chains to empower farmers.



Emerging solutions for crop residue management

- Emerging technologies in crop residue management have strengthened the value proposition for new agri-tech enterprises. A number of in-situ crop residue management practices have been found effective at reducing the issue of post-harvest residue burning, especially in the states of Haryana and Punjab where an estimated 40% of the harvest is burnt as crop residue.¹⁸
- Rice residue – useful for producing manure and ethanol – has emerged as a valuable resource to drive eco-friendly energy systems. In the future, the widespread adoption of these solutions will prove critical in reducing the ill effects of crop burning on public health and farmer economics.¹⁸
- Tools such as the zero tillage drill have demonstrated an ability to make crop management more efficient by removing additional rice residue from fields.¹⁸

¹⁴ APEDA, Ministry of Commerce & Industries et.al. (2022). *Indian Superfood Millets: A \$ 2 billion export opportunity*.

¹⁵ Sattva Consulting. (2023). *Strengthening Millet Value Chain through Private Sector Collaboration*.

¹⁶ P Mishra et.al. (2023). *An Overview of Pulses Production in India: Retrospect and Prospects of the Future Food with an Application of Hybrid Models*.

¹⁷ Dr P Gaur. (2020). *Journal of Food Legumes. Can India Sustain High Growth of Pulses Production*.

¹⁸ S Dhanda. (2022). *Emerging Issues and Potential Opportunities in the Rice-Wheat Cropping System of North-Western India*.



Introduction of disruptive ICT Technologies for quality monitoring, grading and reducing post-harvest losses

- As a part of its Network Program on Precision Agriculture, the ICAR developed nearly 25 fertiliser prescription methods for cereals, pulses, fibre crops and vegetables,¹⁹ which can aid Integrated Nutrient Management.
- Through the National e-Governance Plan Agriculture (NeGP-A) and the 2022-23 Union Budget, the government aims to accelerate the adoption of such advanced digital agri-tech-nologies. Agritech startups are providing drone systems to optimise irrigation systems, IoT based precision agriculture solutions and provide early signs of failure for labour intensive food crops.²⁰



Improving ways of sustainable production

- In recent years, the System of Rice Intensification (SRI) has emerged as a promising technology through knowledge exchange and experimentation between farmers, civil society and agricultural institutes. SRI consists of a set of cultivation practices rooted in water conservation, optimal usage of inputs, wider spacing and active soil aeration. In India, multistakeholder partnerships involving philanthropic foundations, civil society and government have already demonstrated an increase in yields for various paddy varieties in multiple eastern and northern states.
- Given the growth in consumer awareness, organic produce is expected to see a rise in demand in both domestic and international markets. Through the Paramparagat Krishi Vikas Yojana (PKVY), the government has been running participatory guarantee certification of organically-grown agricultural products, providing opportunities for increased organic exports. Moreover, food crops like legumes help increase organic matter and microbial biomass and support sustainable practices like crop rotation.

Challenges



Effects of climate change and natural resource management

Decline in grain yield and quality

- By 2050, climate change is expected to significantly impact the conditions of the Indo-Gangetic Plain which produces approximately 15% of the world's wheat.²¹ Grain quality in terms of protein content and yield are expected to decline for both wheat and rice due to rising temperatures and carbon dioxide levels in the atmosphere.²²
- Wheat production saw a dip of ~2% in 2021 due to the heat waves in Punjab and Haryana.²³

Depleting groundwater resources

- Rice, wheat and sugarcane collectively consume over 80% of India's irrigation supplies. On average, rice and wheat require 5000 and 900 gallons of water respectively which are classified under some of the thirstiest crops.²⁴
- The six major food crop-producing states with unsustainable groundwater levels are Punjab, Rajasthan, Haryana, Tamil Nadu, Gujarat, and Uttar Pradesh.

¹⁹ DARE-ICAR. (2021). *Research Priorities & Technologies for Farmer's Welfare*.

²⁰ V Katekar et al. (2022). *The Application of Drone Technology for Sustainable Agriculture in India*

²¹ V Kaur et al. (2014). *Impact of Climate Change on Wheat and Rice*.

²² J Wang. (2019). *Changes in grain protein and amino acids composition of wheat and rice under short-term increased [CO₂] and temperature of canopy air*.

²³ Ministry of Commerce & Industry. (2022). *Govt protects farmers interest through wheat exports restriction, says APEDA Chairman*

²⁴ World Wide Fund for Nature, *Living Waters. Thirsty Crops*.

Soil health degradation

- Soil degradation is a combination of environmental (e.g. flooding) and social (e.g. decline in per capita land availability, economic pressures etc.) factors which have direct implications in regions like the Indo-Gangetic Plain where wheat and rice are highly dominant.²⁵
- The sustainability of rice production is reported to be at risk due to growing micronutrient deficiencies and arsenic content in the soil.²⁶



Crop management and processing

Plant Protection

- In the Trans-Gangetic Plain, overuse of pesticides is a common practice, affecting the health of crops such as rice. This is further exacerbated by the inadequate use of organic matter and poor crop cycle planning.²⁶
- Although policy interventions related to plant protection and quarantine have helped train farmers, there is still a need to focus on the sensitisation and capacity building of private players and the larger ecosystem involved in promoting best practices to prepare for pest attacks.²⁶

Farm mechanisation

- Research and development (R&D) programmes have disproportionately focused on the wheat-rice cropping system. The comparatively lesser focus on innovation in equipment has led to insufficient diversification in the cultivation of food crops.²⁷
- Private brands face issues like inefficient pre-processing, cleaning and grading of major millet varieties at the farm level which result in undesirable and inconsistent quality during procurement.¹⁵

Gaps in agricultural extension

- In the case of organic and natural farming, state officials involved in extension services have an inadequate understanding of farming practices because of curriculum gaps. This limited know-how fails to convert farmers attuned to cultivating crops with chemical inputs for high productivity to more sustainable agricultural practices.²⁸



Post harvest and market challenges

Gaps in the millet value chain

- Despite an increase in production, millet and millet-based products occupy an average share of 1-5% in the manufacturing basket of major food brands.¹⁵ Farmers are often constrained by a limited understanding of market needs pertaining to grain quality and processing requirements. Moreover, private brands fail to procure directly from farmers due to the associated financial costs and a lack of trust in the quality of the produce.
- The lack of awareness among consumers regarding the nutritional value of certain high-value coarse cereals results in low demand, thus disincentivising farmers.²⁰
- Despite the active promotion of millets in states like Odisha, logistical challenges cause delays in the transportation of produce from the producer to the consumer.²⁹ As a majority of farmers fall in the small and marginal categories, aggregation platforms within a radius of 5 to 6 km from farm-gate can facilitate direct linkages with local retail and terminal markets.

25 R Bhattacharya et al. (2015). *Soil Degradation in India: Challenges and Potential Solutions*.

26 R.Chand et al.,(eds.), *India Studies in Business and Economics. (2022). Indian Agriculture towards 2030*.

27 B Bhandari. (2023). *Making India a Global Power House in the Farm Machinery Industry*.

28 R Nandi. (2019). *Agriculture Extension System in India: A Meta-analysis*.

29 S Das. (2020). *Supply Chain of Millets: An FPO perspective. (With a special reference to Odisha)*.

Storage and Residue Management

- The surplus production of certain food crops remains a significant challenge; for instance, the Food Corporation of India's rice stocks amounted to 21 metric tonnes in January 2020 as opposed to the buffer requirement of 7 metric tonnes.³⁰ As per the Committee on Doubling Farmer's Income, upgradation of existing godowns and warehouses need to be prioritised and complemented with District / State wise storage plans to ensure efficient spatial distribution.
- On-farm burning of wheat and rice crop residue has intensified in recent years due to the use of mechanised harvesters and the high cost of labour involved in removing crop residues.³¹
- Crops like processed millets have limited shelf lives, presenting storage and procurement problems for private players.¹⁵

Ministries and Departments



30 Trade Promotion Council of India. (2020). *Foodgrain storage losses in India*. Retrieved July 2023.

31 Ministry of Agriculture & Farmers Welfare. (2022). *Promotion of Agricultural Mechanisation for In-situ Management of Crop Residue in the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi*.

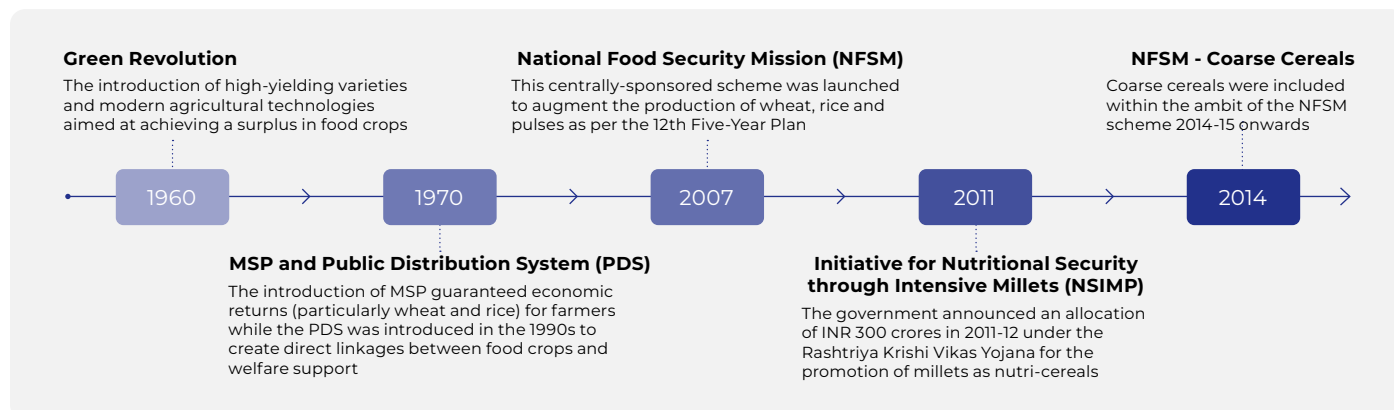
32 Ministry of Agriculture and Farmer Welfare. Retrieved on July 2023.

33 National Bank for Agriculture and Rural Development. (2022). *Food and Nutritional Security in India: Charting the way to a robust Agri-food system*.

34 Ministry of Consumer Affairs, Food & Public Distribution. Retrieved on July 2023.

Policy evolution over the years

While food crops have always prominently contributed to food security, a deliberate shift towards sustainability and the promotion of eco-friendly and climate-resilient food crops has occurred.



Several agricultural policies have aimed to address the challenges associated with the sustainable production of food crops



EFFECTS OF CLIMATE CHANGE AND NATURAL RESOURCE MANAGEMENT

Pradhan Mantri Krishi Sinchayee Yojana

Improve farm water management through precision agriculture

Initiative for Nutritional Security through Intensive Millets Promotion

Promote millet cultivation, processing and value addition

National Mission for Sustainable Agriculture

Anchor multiple missions to foster climate-smart agriculture for all major food crops

Paramparagat Krishi Vikas Yojana

Adopt organic practices, such as the System of Rice Intensification, to produce crops

National Crop Insurance Scheme

Provide financial support to farmers in the event of crop failure due to calamities, pests and diseases

National Mission on Natural Farming

Promote natural farming methods to cultivate food crops



CROP MANAGEMENT AND PROCESSING

National Mission on Agricultural Extension & Technology

Restructure agricultural extension services to deploy new productivity-improving technologies

National Food Security Mission

Enhance the production of all major cereals and pulses across the 28 states of India

Targeting Rice-fallow Areas

An NFSM programme to promote the cultivation of pulses in rice-fallow areas

PM Formalisation of Micro Food Processing Enterprises

Enhance the competitiveness of existing unorganised food processing micro-enterprises

Rashtriya Krishi Vikas Yojana

Promote the integrated development of all major food crops, and augment infrastructure, storage and warehousing

Sub-mission on Plant Protection and Plant Quarantine

Encourage the appropriate use of biopesticides and integrated pest management practices for all crops



POST-HARVEST AND MARKET SUPPORT

Minimum Support Price

Ensure guaranteed procurement price for the direct purchase of food crops from farmers

Integrated Scheme on Agricultural Marketing

Provide farmers with better access to competitive markets and develop infrastructure for storage and other agribusiness operations

Online Storage Management

Offer a single source of information regarding the food grains stored in the country

Integrated Scheme on Agricultural Cooperation

Provide financial assistance to cooperatives for the marketing, storage and processing of agricultural produce

Open Market Sale Scheme

An FCI programme to facilitate the sale of surplus food grains, primarily wheat and rice

Agriculture Infrastructure Fund

Supply medium- to long-term debt financing to support projects aimed at improving post-harvest management

National e-Governance Plan in Agriculture

Digitise agricultural operations and use ICT to access information promptly

Digital Agriculture Mission

Support cutting-edge farming technologies such as remote sensing, geographic information system (GIS), drones etc.

Essential Commodities Act, 1955

Protect the availability of certain essential commodities (which include all food crops) from supply disruptions

Ethanol Blended Petrol Programme

Leverage sources of alternative and eco-friendly fuels such as wheat and sugar to reduce energy import dependency

Agri Stack

Facilitate access to finance, inputs and market information through a federated farmers' database



Way Forward

 Shift towards diversification and climate resilient crops	Intercropping and new varieties are likely to dominate India's cultivation preferences • The NFSM has been promoting the intercropping of pulses with coarse cereals, oilseeds and commercial crops such as cotton for better use of land and water resources.³⁵ • In 2021, the Prime Minister of India released 35 varieties of climate-resilient food crops developed by the ICAR to address challenges related to climate change and malnutrition. • Strategic breeding to develop tolerance for heat and moisture stressors will be critical for wheat and rice cultivation to ensure yield maximisation during changing climates. Other crops like climate resilient maize varieties have been successfully deployed in Sub Saharan Africa and can support diversification efforts in India.³⁶	Mainstreaming climate-resilient crops will enable long-term food security • Several hybrid varieties of major millets like Sorghum and Bajra exist today but their scalability depends on developing seed quality and productivity in all local conditions.³⁷ Moreover, introducing legumes into organic production can support nitrogen fixation and improve India's soil fertility for long term.³⁸ • The strengthening of supply chains in terms of quality and availability for the private sector will be critical for FMCG brands to unlock the value of advantageous cereals like millets.¹⁵ • State Millet Missions in Odisha and Madhya Pradesh are improving the market access for millets and millet products by linking them with the Integrated Child Development Schemes and livelihood schemes²⁶
 Water and Soil Management	Optimising the usage of essential resources can improve sustainability • For water-intensive crops like rice, the government is testing and integrating conservation technologies such as the System of Rice Intensification and direct seeding of rice. • Through flagship programmes like the National Mission for Sustainable Agriculture, the government has promoted natural farming, nutrient-based subsidies, and soil testing labs, particularly for cereals and pulses.	Reducing the dominance of the wheat-rice system in vulnerable regions can promote sustainability • With campaigns like Sahi Fasal, there is a growing focus on nudging farmers to grow less water-intensive crops that are more suitable for agroecology.³⁹ • The government is encouraging intercropping and diversification to reduce dependency on input-intensive cereals. Assistance is being provided through demonstrations on the latest crop production technologies, and integrated nutrient and pest management to protect soil health and resources.³²
 Augmentation of Food-processing Methods	Biofortification along with higher adoption of nutri-cereals and pulses can lead to improved nutrition • The distribution of biofortified rice and millets at subsidised prices through the PDS is a cost-effective way to reach nutrition-deficient groups.²⁶ • In 2021, the ICAR released high-zinc rice, protein-enriched wheat, hybrid maize varieties and iron-rich millets for cultivation.	Processing is integral to the development of the millet value chain in India • Private sector participation can help improve the quality of millet products through modern food processing techniques and value addition.¹⁵ • Major food brands and even small-scale startups have already launched millet-based products, mainly incorporating Sorghum (Jowar), pearl millet (Bajra) and finger millet (Ragi).¹⁵

35 Ministry of Agriculture & Farmers Welfare. (2018). *National Food Security Mission- Operational Guidelines*.

36 J E Cairns & B M Prasanna. (2018). *Developing and deploying climate-resilient maize varieties in the developing world*.

37 NITI Aayog. (2023). *Promoting Millets in Diets, Best Practices across States / UTs of India*.

38 A Singh & H S Chahal. (2020). *Organic Grain Legumes in India: Potential Production Strategies, Perspective, and Relevance*.

39 Department of Agriculture & Farmers Welfare. (2020). *Annual Report*.

 Mechanisation & Technology	Farm mechanisation is expected to grow and compensate for lack of labour in production • Under various farm mechanisation programmes including NFSM, the government has promoted custom hiring centres that allow renting of both large equipments and power operated tools.⁴⁰ These centres play a crucial role in offsetting adverse economies of scale due to small and fragmented landholdings. • In rice cultivation, several crucial operations such as transplanting and harvesting are done manually. However, there is a growing adoption of artificial intelligence, remote monitoring, and rice transplanters to manage labour intensive aspects of production.⁴⁰	Technology is expected to play a vital role in driving optimisation and efficiency • There is a growing focus on optimising inputs such as fertilisers, pesticides and water through precision farming for resource-intensive food crops. Since 2023, the government has initiated subsidies for procurement of drones to foster adoption of precision agriculture by small farmers at the last mile.
 Price and market based support	Policy reforms on subsidy support to farmers provide added support • Spending on subsidies has been on the rise despite enormous fiscal constraints. • Rationalisation of various input subsidies (e.g. fertilisers, electricity etc.) will play a critical role in enabling a shift towards a diverse crop pattern for cereals and pulses through interventions in precision farming and micro-irrigation. There is an emerging focus on reducing dependency on external inputs for farming.²⁶	There has been a diversification of the public procurement basket for pulses and nutri-cereals • Several states have taken promising steps to include nutri-cereals and pulses in their procurement basket.²⁶ • Between 2007-2019, the share of pulses in public procurement grew by 5% while that for rice dropped by 33% to allow for a broader basket.
 Post Harvest Management	Policy attention towards primary processing and value-addition-based support has increased • Primary processing units and small storage bins were included in the scope of the NFSM from 2020-21 onwards.³² • Through the ICAR's Farmers First Programme, the Ministry of Agriculture & Farmers' Welfare (MoAFW) supports farmers and young entrepreneurs with the cultivation of value-added millet food products.⁴¹	Storage facilities and efficient procurement processes are crucial for ensuring quality and productivity • Agri-tech startups and private players have been augmenting the efforts of Food Corporation of India and other central warehouses to significantly reduce post-harvest loss and improve the shelf lives of cereals like millets.⁹

40 PK Guru. (2018). *Rice Mechanisation in India*.

41 ICAR, Farmer's First Programme. Retrieved July 2023.



Gender-inclusive Value Chains

Modern technologies are expected to reduce drudgery for women farmers

- Due to limited non-farm opportunities in rural regions of India, agriculture still remains a major source of employment for women (76.2% as per PLFS 2022-23).
- Women perform multiple physically demanding tasks for cultivation of food crops such as preparing seedbeds, transplanting and sowing. Farm mechanisation and technologies like System of Rice Intensification are significantly reducing workload of women through efficient methods.⁴²

Increasing the productive and entrepreneurial capacity of rural women

- Growing migration of men due to climate stressors is likely to increase feminisation of agriculture. Institutions like women FPOs and SHGs can help strengthen the linkages between women farmers and other value chain actors.⁴³
- For both cereals and pulses, women farmers can spearhead grassroots movements in sustainable practices. However, it will be important to increase their control over means of production with access to market and skill development.⁴⁴

Recommended Readings

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⁴² Council on Energy, Environment and Water. *System of Rice Intensification in India*. Retrieved on October 2023.

⁴³ Manage. (2020). *Success Stories on Women Farmer Producer Organisations*.

⁴⁴ Food and Agriculture Organisation. (2016). *Developing Gender Sensitive Value Chains- A Guiding Framework*

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